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SCD - BADMINTON STORE WEB APP
Software Architecture Document

Version 1.1

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Revision History

Date	Version	Description	Author
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		Write the Architectural Goals and Constraints	Nguyễn Minh Kha
		Use-case Model	Lê Hồ Đan Anh
		Logical view: 4.1 Component: Service	Nguyễn Trần Thiên Phú
		Logical view: 4.2 Component: UI Components	Nguyễn Minh Kha
		Logical view: 4.3 Component: Controller (Frontend)	Võ Nhật Hào
		Logical view: 4.4 Component: Controller (Backend)	Nguyễn Ngọc Mai Xuân
		Logical view: 4.5 Component: Router	Võ Nhật Hào
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Software Architecture Document

1. Introduction

1.1 Purpose

This Software Architecture Document's primary goal is to illustrate the architectural decisions and rules that underpin the SCD Badminton. It serves as a reference for stakeholders, developers, designers, and testers who are involved in the software development process. This document builds cooperation, reduces ambiguity, and ensures that the development team conforms to a single view and design standards by outlining the system's architecture.

1.2 Scope

The general design of the SCD Badminton, including its key components, modules, interfaces, and interactions, is covered in this Software Architecture Document. The scope of this document covers:

- Customer/Guest-facing Website: Where customers browse products, search, add items to the cart, place orders, and complete payments.
- Admin dashboard: Where administrators manage products, categories, orders, customers, and other website content.

1.3 References

List of documents used to inform the creation of this Software Architecture Document:

- Vision Document
- Use-case Specification

1.4 Overview

The following sections present a detailed examination of the architecture of the SCD Badminton website. The document is structured as follows:

- Introduction
- Architecture Goals and Constraints
- Use-case Model
- Logical View
- Deployment
- Implementation View

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2. Architectural Goals and Constraints

This section outlines the architectural goals and constraints that significantly influenced the structure, design decisions, and development process of the SCD - Badminton Store Web Application. These considerations address both functional and non-functional requirements that directly impact performance, security, maintainability, and user experience. These identified goals and constraints provide the foundation for ensuring scalability, security compliance, and robust e-commerce capabilities, appropriate for the sports retail industry.

2.1 Architectural Goals

- **Performance optimization:** Each button or feature on the website such as product filters, “Thêm vào giỏ hàng” hay “Mua ngay” must respond within 1 second. Image loading should be lazy-loaded for optimized performance on both desktop and mobile platforms.
- **System scalability:** The system should be capable of handling thousands of product entries, including different categories, variants and pricing schemes. This supports future growth without the need for a major architectural redesign.
- **Access control & session security:** Users must be authenticated via JWT-based login sessions. Each session token has a 1-day expiration, with role-based access control restricting unauthorized access to admin pages and sensitive routes.
- **Data integrity & input validation:** All users input must be validated on both client and server side to prevent injection attacks, malformed data and broken transactions.
- **Confidential & privacy:** Passwords must be hashed using the **Bcrypt** algorithm before stored in the MongoDB database. Sensitive customer data such as email, order history, ... must be protected against unauthorized access.
- **Promotion management:** The system must support dynamic promotional campaigns, including percentage-based discounts, direct discounts, flash sales and featured product highlights. Admin should be able to active/ deactivate promotions in real-time.
- **Responsive user interface:** The frontend should be responsive and optimized for mobile devices. All critical features such as search, product selection, cart and checkout must be fully accessible and touch-friendly.

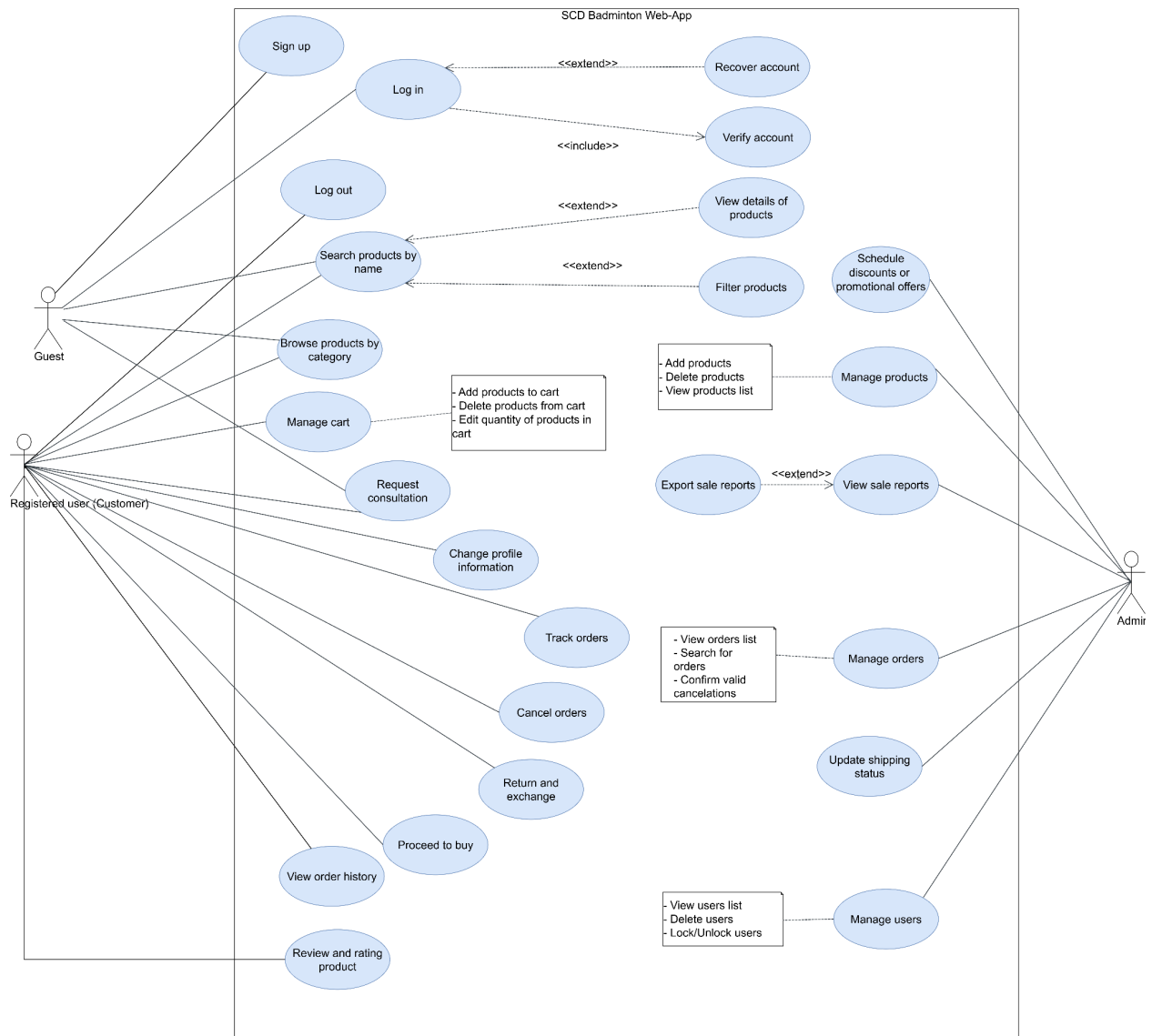
2.2 Architectural Constraints

- **Platform:** Web-based eCommerce application
- **Applicable Standards:** The application must be compatible with modern browsers on both Windows and macOS. Mobile browsers must also be supported.
- **Programming Languages:** HTML, CSS and JavaScript are used throughout the application. Data logic is handled via NoSQL.
- **Frameworks & Libraries:**

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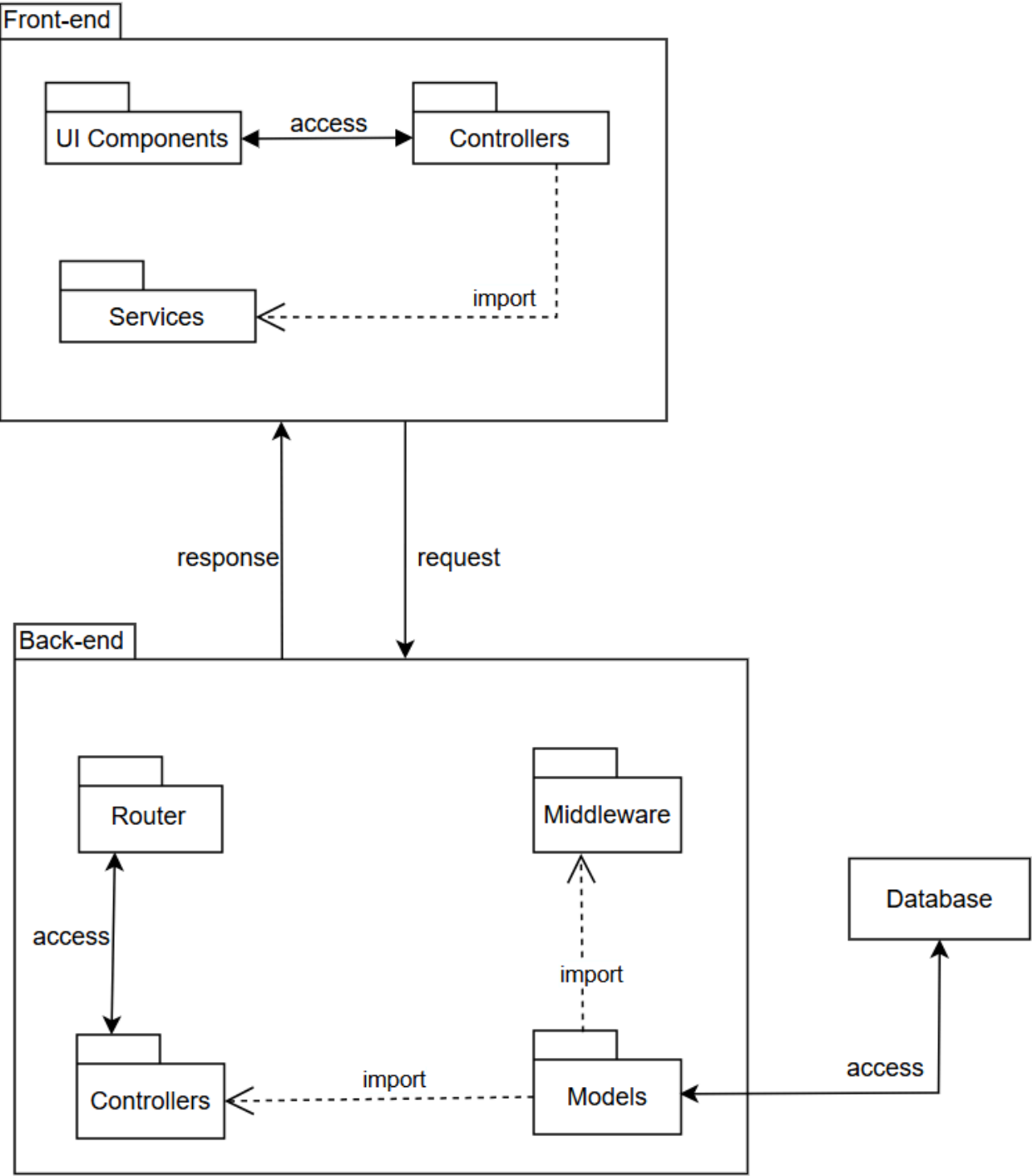
- **Frontend:** ReactJS with pure CSS
- **Backend:** ExpressJS on NodeJS with RESTful API architecture
- **Database:** MongoDB with schema definition via Mongoose
- **Limited Integration with third-party Services:** For simplicity and manageability, third-party services are limited to image upload via Cloudinary and address lookup via local Vietnamese province APIs.

3. Use-Case Model



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4. Logical View



The web application is built following the MERN stack architecture and adheres to the MVC (Model -

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View - Controller) design pattern across both frontend and backend components.

Frontend: The frontend is designed with an emphasis on responsiveness and visual appeal. It utilizes ReactJS to manage UI components and state logic.

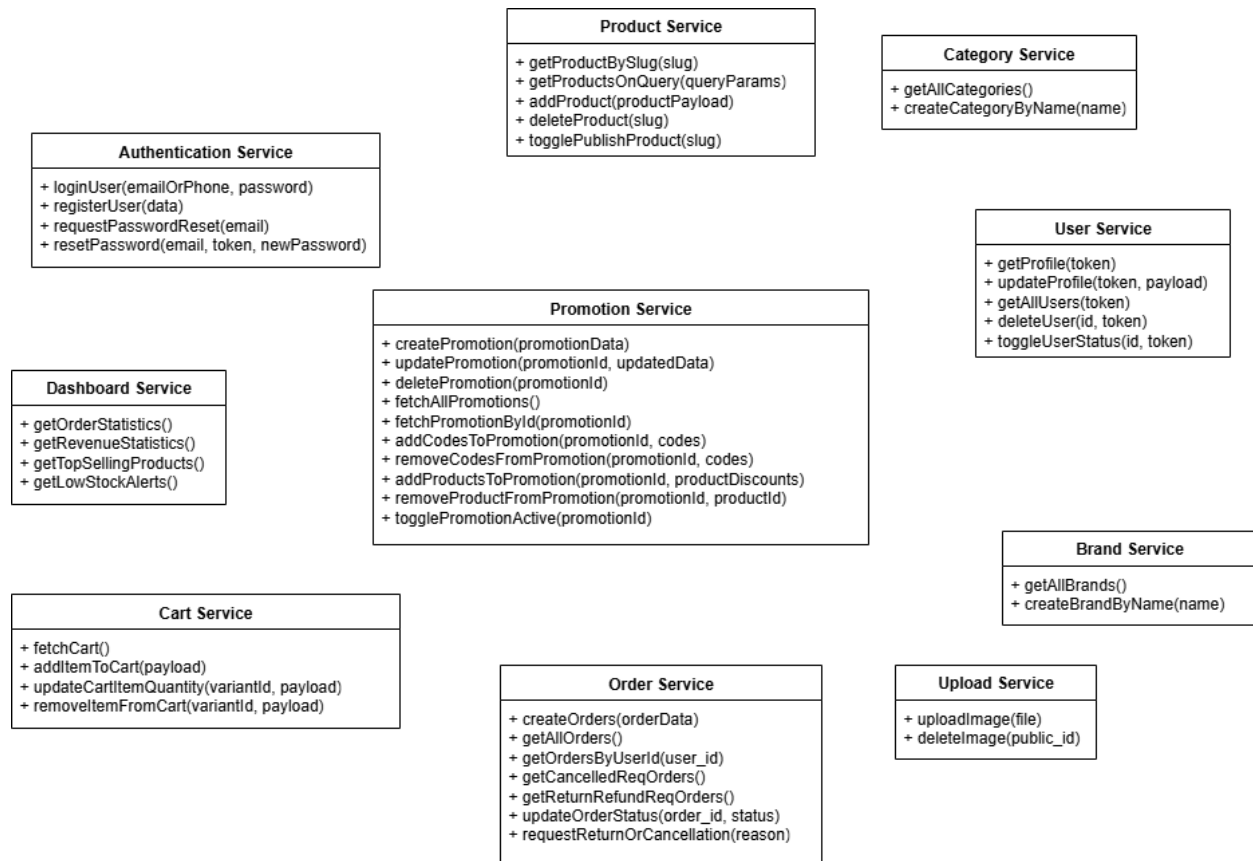
- **Models (Local Data):** Include both static data stored on the server and dynamic data fetched from the backend.
- **Views:** These are React components responsible for rendering the user interface and presenting data to the user.
- **Services:** Facilitate communication with the backend by sending API requests, processing responses, and managing data flow between the UI and the server.
- **Controllers:** Handle user interactions and coordinate logic by managing state changes or triggering services.

Backend: The backend is developed using Node.js with the Express.js framework, offering a robust platform for server-side logic, routing, and middleware integration. It functions as a RESTful API server, handling client requests and returning appropriate responses.

- **Models:** Defined using Mongoose to structure and validate data in MongoDB.
- **Controllers:** Manage application logic, interact with models, and return appropriate responses to client requests.
- **Middleware:** Handle cross-cutting concerns such as authentication, input validation, and error handling.
- **Routing:** Maps incoming API requests to their respective controllers, ensuring the correct functionality is executed based on the endpoint.

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4.1 Component: Services



Authentication Service: This service is responsible for all user authentication processes. It manages user sign-in, registration, and password recovery.

- **loginUser(emailOrPhone, password):** Verifies user credentials and logs them in.
- **registerUser(data):** Creates a new user account.
- **requestPasswordReset(email):** Initiates the password reset process for a given email.
- **resetPassword(email, token, newPassword):** Completes the password reset process using a verification token.

User Service: This service manages user data and administrative actions related to users. It handles user profiles and status management.

- **getProfile(token):** Retrieves the profile information of the currently logged-in user.
- **updateProfile(token, payload):** Updates the profile information of the logged-in user.
- **getAllUsers(token):** Fetches a list of all users in the system (admin access required).
- **deleteUser(id, token):** Deletes a user account (admin access required).

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- **toggleUserStatus(id, token):** Changes a user's status.

Product Service: This service handles all operations related to products. It manages the product catalog, including creation, retrieval, and updates.

- **getProductBySlug(slug):** Retrieves detailed information for a single product using its URL-friendly slug.
- **getProductsOnQuery(queryParams):** Fetches a list of products based on filter or search parameters.
- **addProduct(productPayload):** Adds a new product to the catalog.
- **deleteProduct(slug):** Removes a product from the catalog.
- **togglePublishProduct(slug):** Changes the visibility of a product on the website (e.g., makes it live or hidden).

Category Service: This service manages the categorization of products.

- **getAllCategories():** Retrieves a list of all available product categories.
- **createCategoryByName(name):** Creates a new product category.

Brand Service: This service manages product brand information.

- **getAllBrands():** Retrieves a list of all available product brands.
- **createBrandByName(name):** Creates a new product brand.

Cart Service: This service is responsible for managing the user's shopping cart.

- **fetchCart():** Retrieves the contents of the user's current shopping cart.
- **addItemToCart(payload):** Adds a new item to the cart.
- **updateCartItemQuantity(variantId, payload):** Changes the quantity of an existing item in the cart.
- **removeItemFromCart(variantId, payload):** Removes an item from the cart.

Order Service: This service manages all aspects of customer orders after the checkout process.

- **createOrders(orderData):** Creates a new order from the items in the cart.
- **getAllOrders():** Retrieves a list of all orders in the system (for admin use).
- **getOrdersByUserId(user_id):** Retrieves the order history for a specific user.
- **getCancelledReqOrders():** Fetches lists of orders with pending cancellation requests.
- **getReturnRefundReqOrders():** Fetches lists of orders with pending return/refund requests.
- **updateOrderStatus(order_id, status):** Updates the status of an order (e.g., "shipped," "delivered").
- **requestReturnOrCancellation(reason):** Allows a user to request a return or cancellation for an order.

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Promotion Service: This service handles the creation and management of promotional campaigns, discounts, and promotional codes. It provides full CRUD (Create, Read, Update, Delete) functionality for promotions. Specifically:

- **createPromotion(promotionData):** Creates a new promotional campaign with the provided promotional data.
- **updatePromotion(promotionId, updatedData):** Updates an existing promotion with new data based on its ID.
- **deletePromotion(promotionId):** Deletes a promotion based on its ID.
- **fetchAllPromotions():** Retrieves a list of all existing promotions.
- **fetchPromotionById(promotionId):** Fetches the details of a specific promotion based on its ID.
- **addCodesToPromotion(promotionId, codes):** Adds promotional codes to a promotion.
- **removeCodesFromPromotion(promotionId, codes):** Removes promotional codes from a promotion.
- **addProductsToPromotion(promotionId, productDiscounts):** Links or adds products with specific discounts to a promotion.
- **removeProductFromPromotion(promotionId, productId):** Removes a product from a promotion.
- **togglePromotionActive(promotionId):** Activates or deactivates a promotion.

Dashboard Service: This service provides aggregated data for an administrative dashboard, offering insights into the store's performance.

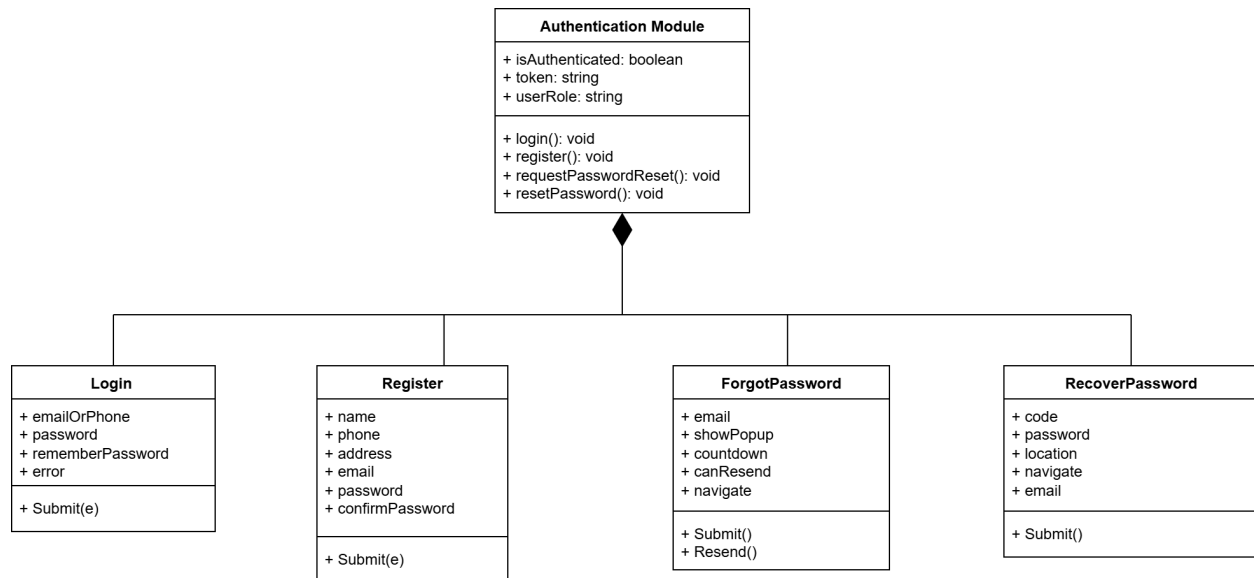
- **getOrderStatistics():** Gather statistics related to orders (e.g., number of orders over state).
- **getRevenueStatistics():** Gather statistics related to sales revenue.
- **getTopSellingProducts():** Returns a list of the best-selling products.
- **getLowStockAlerts():** Identifies products that are running low on inventory.

Upload Service: This is a utility service for handling image file uploads.

- **uploadImage(file):** Uploads an image file and returns its identifier.
- **deleteImage(public_id):** Deletes an uploaded image using its public ID.

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4.2 Component: UI Components



The Authentication module is responsible for verifying user identities and ensuring secure system-wide access. It handles the processes for users to sign up, log in, and recover their credentials. Furthermore, it enforces access permissions based on a user's assigned role.

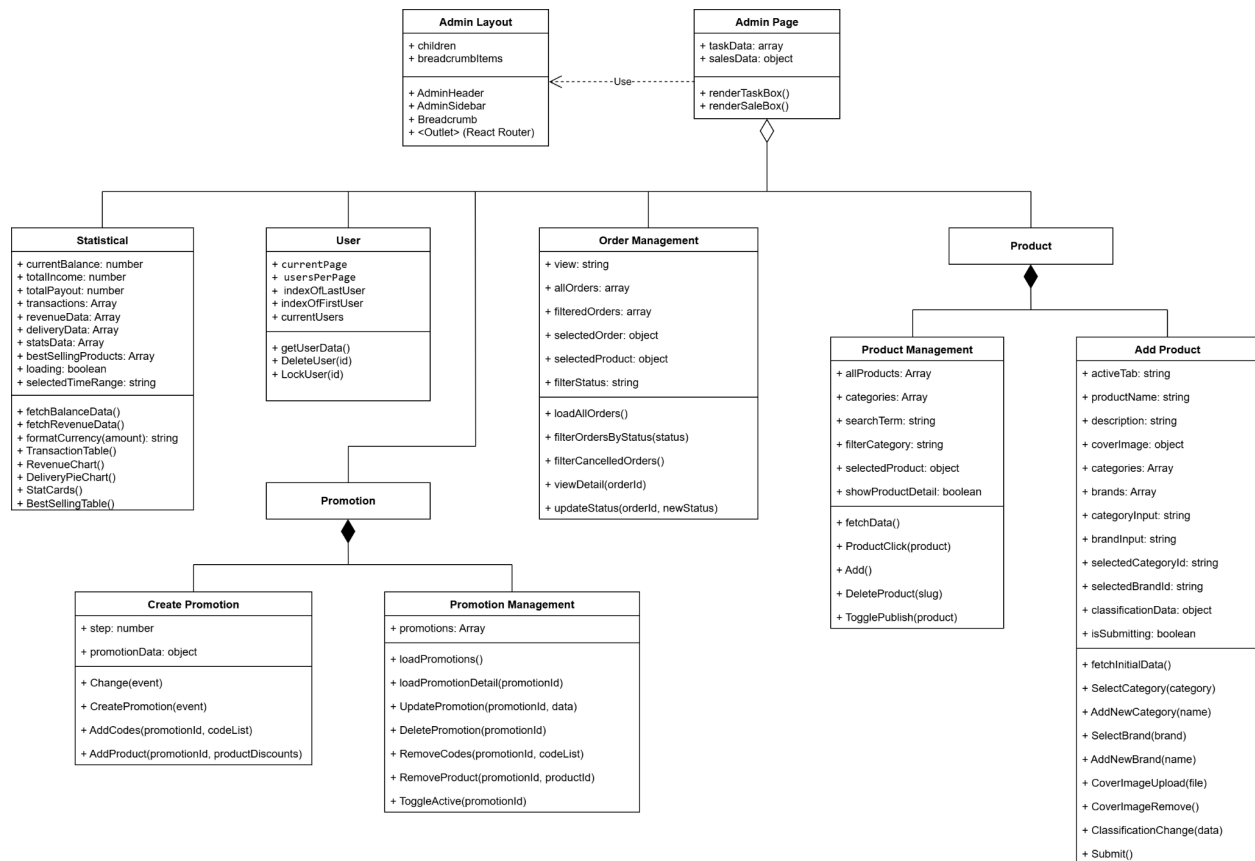
Authentication:

- **Login:** Existing users sign in using email and password. Successful login returns a JWT token stored locally for session management.
- **Register:** New customers can create an account via a validated form.
- **Forgot Password:** Provides a technique for users to recover their password if forgotten.
- **Recover Password:** Allows users to reset their password using the received token.

Access Control:

- **Authenticated users** (Admin or Logged-in User) can access features based on their role.
- **Guests** can only access public pages, perform some functions such as searching for products, viewing product details but will be limited in personalization or transaction features.

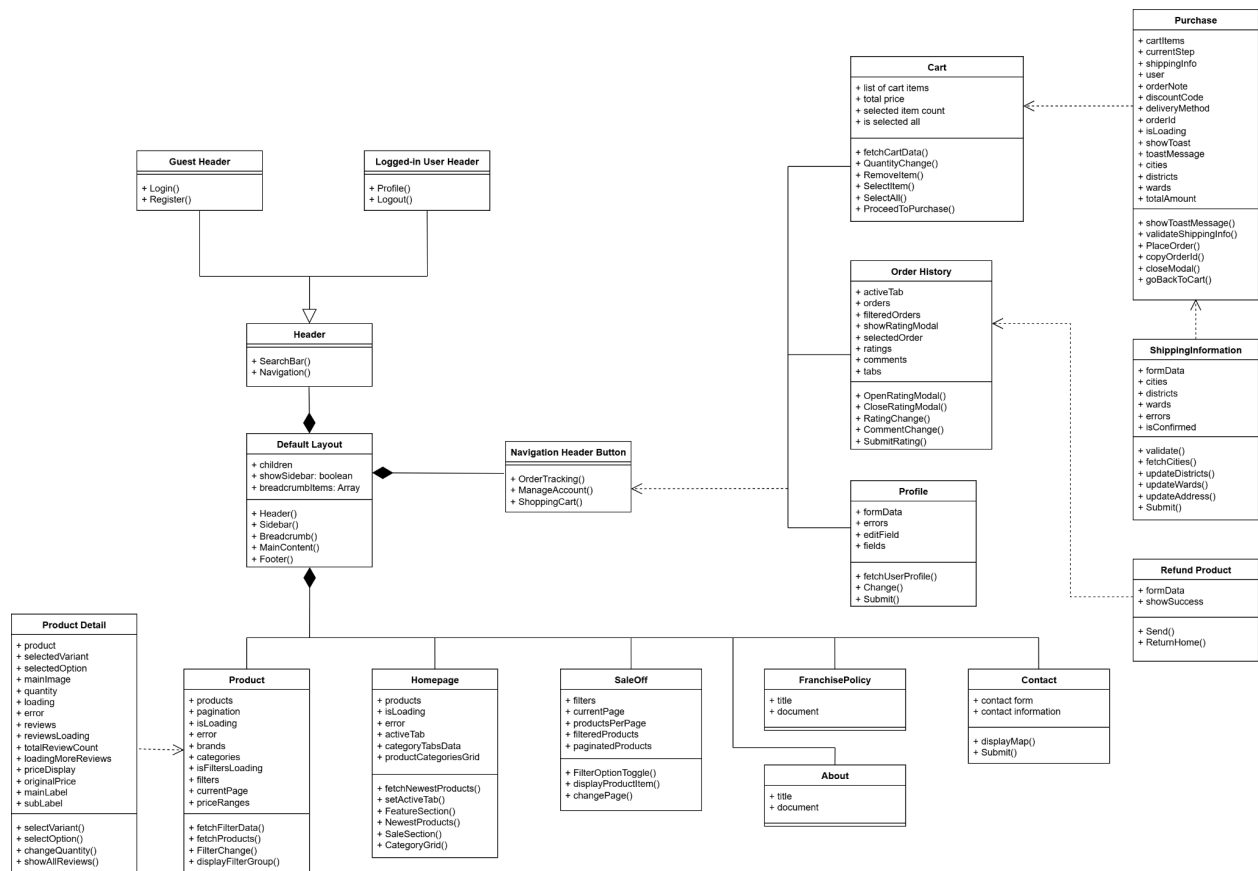
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Administrator:

- **User:** Admin can manage registered user accounts. This includes taking actions such as disabling or banning accounts that violate platform policies.
- **Product:** Admin can manage the entire product catalog. This includes performing operations such as viewing product lists, filtering by category, checking product details, adding new products, and removing obsolete or discontinued products.
- **Order Management:** Admin can monitor all customer orders. They can update order status, confirm successful delivery, and handle customer-submitted issues related to specific transactions.
- **Promotion:** Admin can create and manage promotional campaigns. This involves defining discount rules, associating products with campaigns, generating coupon codes, and enabling or disabling active promotions.
- **Statistical:** Admin can view analytical dashboards providing insights into sales performance, user activity, top-selling products, and revenue trends. This empowers data-driven decision-making to optimize store operations.

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Guest: Allows non-logged-in users to access general content and features.

- **HomePage:** The landing page showcasing featured products, promotions, and general brand content.
- **Product:** Displays a list of products available on the platform, including filtering and basic information.
- **Sale Off:** Showcases products that are currently part of promotional campaigns or discounted sales.
- **Franchise Policy:** Provides information about the platform's franchising opportunities and business model.
- **About:** Introduces the brand's mission, vision, and key highlights.
- **Contact:** Offers users a way to reach out via contact form or support information.
- **Product Detail:** Displays detailed product information including price, variants, and user reviews.

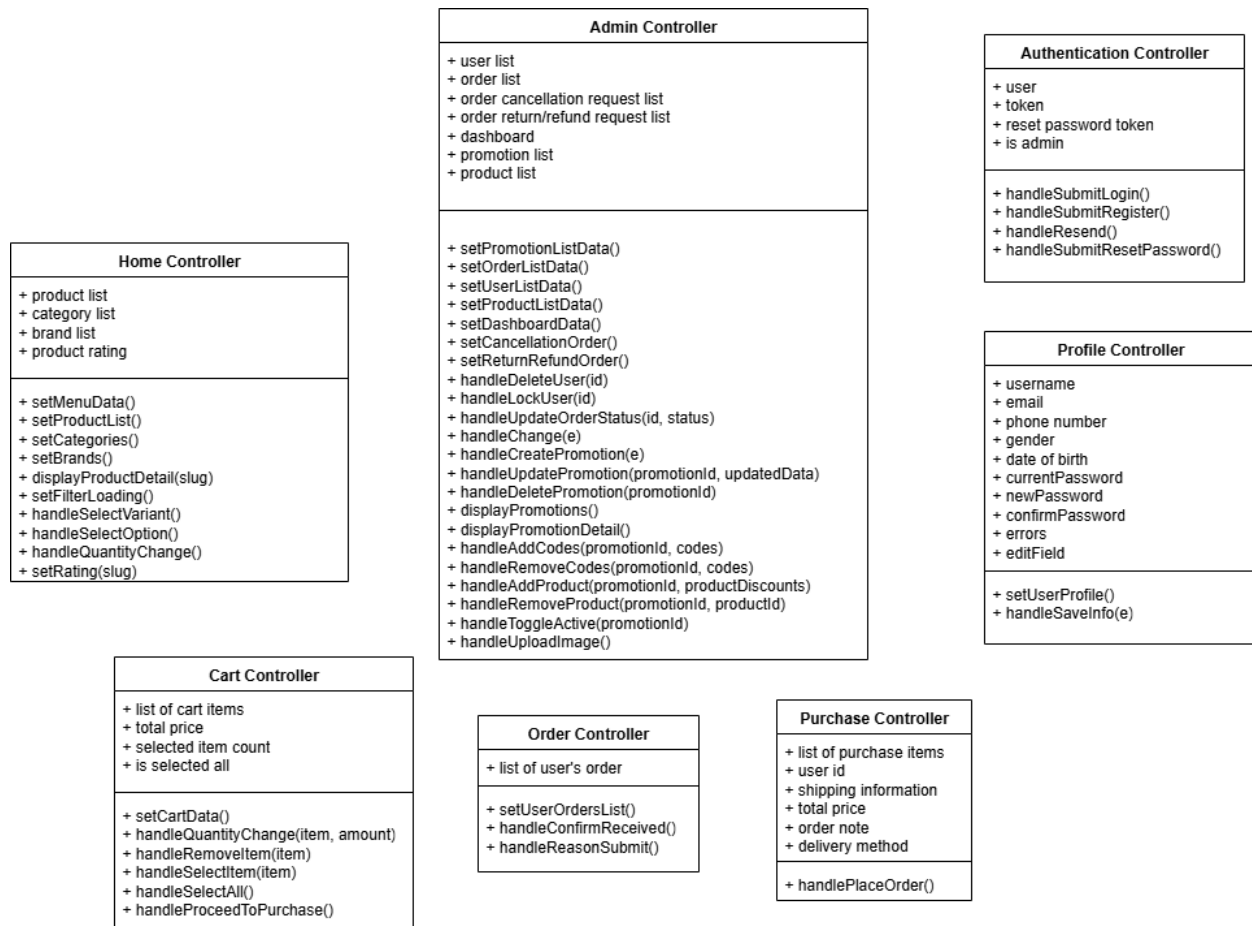
Logged-in User: Authenticated users will be able to perform all actions that **Guests** can perform, along with access to shopping, order management, cart management, profiles, and transactions within the platform.

- **Profile:** Shows the authenticated user's account information and editable preferences.
- **Cart:** Displays all selected products and variants, allows to modify quantity, delete products and purchase.
- **Shipping Information:** Allows users to enter and save shipping information, select shipping methods, and leave order notes.
- **Purchase:** Final confirmation page to review order summary, total amount.

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- **Order History:** Lists previous purchases with status tracking, order details, and optional product rating.
- **Refund Product:** Provides an interface to submit refund or return requests for eligible purchased items.

4.3 Component: Controllers (Frontend)



Admin Controller: This component is responsible for managing admin pages:

- Manage users (view, lock/unlock, delete, update status).
- Manage orders, including cancellation and refund requests.
- Control promotional content (create, edit, delete, toggle active promotions).
- Handle uploading and managing product data.
- Generate dashboard data and reports from order data.

Authentication Controller: Handles user identity verification and access control:

- Manage user login and registration workflows.
- Handle token-based authentication.
- Provide functionality for password reset.

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Profile Controller: Responsible for managing the user's personal profile:

- Display and edit user information (email, phone, date of birth, etc.).
- Allow users to update their passwords securely.
- Enable users to save profile changes.

Home Controller: Manages data displayed on the main storefront or landing page:

- Load category-specific product list, brands.
- Handle product detail, ratings views and filtering options.
- Support selection of product variants, options, and user interactions like rating or quantity selection.

Cart Controller: Handles all shopping cart-related functionality:

- Display current items in the cart with total price.
- Manage item quantities, removal, and selection.
- Support checkout preparation.

Order Controller: Responsible for order confirmation and history:

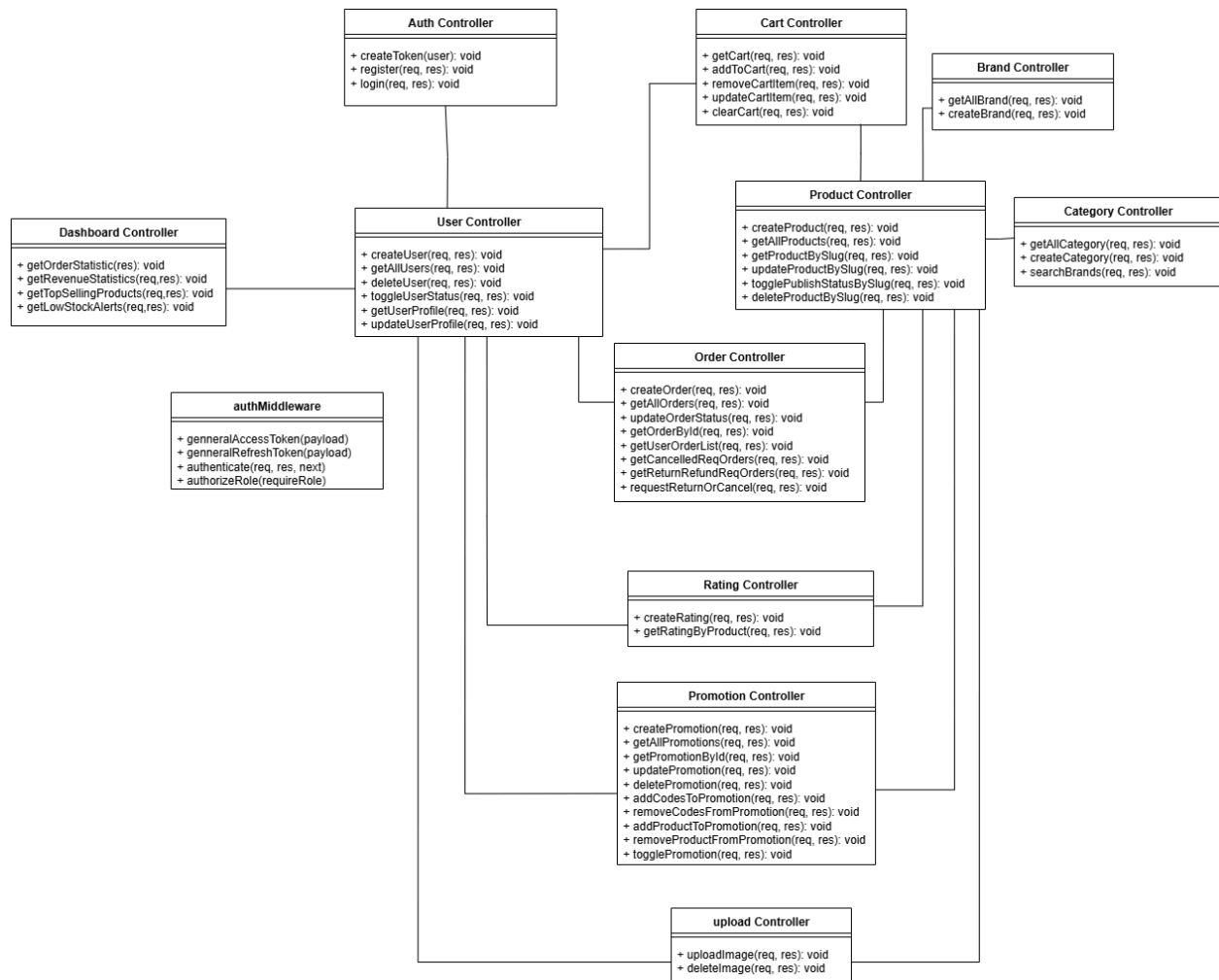
- Load the user's order list.
- Confirm status of delivered items.
- Enable users to return/refund or review previous orders.

Purchase Controller: Manages the purchase process and order placement:

- Store information about purchased items, shipping, order notes, methods, and pricing.
- Facilitate final order placement and submission.

4.4 Component: Controllers (Backend)

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Auth Controller: Manages user authentication and registration. This includes actions such as login, registration, and token creation.

- Create token: Generate a token for the user after authentication.
- Register user: Manage the registration of a new user account.
- Login: Handle the login request for existing users.

User Controller: Manages user information, including actions like creating, updating user details, and updating user status.

- Create user: Create a new user account.
- Get user information: Retrieve information about a specific user or all users.
- Update user information: Update the user's profile information.
- Update user status: Update the user's status (e.g., activate, deactivate).

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Product Controller: Manages products within the system. This includes creating, retrieving, and updating product information.

- Create product: Add a new product to the system.
- Get product information: Retrieve a list or detailed information of products.
- Update product: Update product information or its publication status.

Cart Controller: Manages the user's shopping cart, including actions like adding, removing, and updating items in the cart.

- Get cart: Retrieve the user's shopping cart information.
- Add product to cart: Add a product to the user's cart.
- Update cart: Update the quantity or items in the cart.

Brand Controller: Manages product brands. This includes actions like retrieving the list of brands and creating new brands.

- Get brand: Retrieve a list of available brands.
- Create brand: Add a new brand to the system.

Category Controller: Manages product categories. This includes actions like creating categories and searching for brands within categories.

- Get category: Retrieve a list of all categories.
- Create category: Add a new category to the system.

Order Controller: Manages user orders, including actions like creating orders and updating order status.

- Create order: Create a new order for the user.
- Update order status: Update the status of an order (e.g., processing, completed).
- Get user's order list
- Get all orders, cancellation request orders, return/refund request orders for admin
- Get order's detail
- Handle accept cancellation request orders, return/refund request orders.

Rating Controller: Manages product ratings given by users. This includes actions like creating new ratings and retrieving product ratings.

- Create rating: Add a new rating for a product.

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- Get product rating: Retrieve a list of ratings for a product.

Promotion Controller: Manages promotional programs. This includes creating, updating, and deleting promotional programs.

- Create promotion: Create a new promotion for products.
- Update promotion: Update the details of an existing promotion.
- Delete promotion: Delete expired or inactive promotions.

Upload Controller: Manages the uploading of images for products or user profiles.

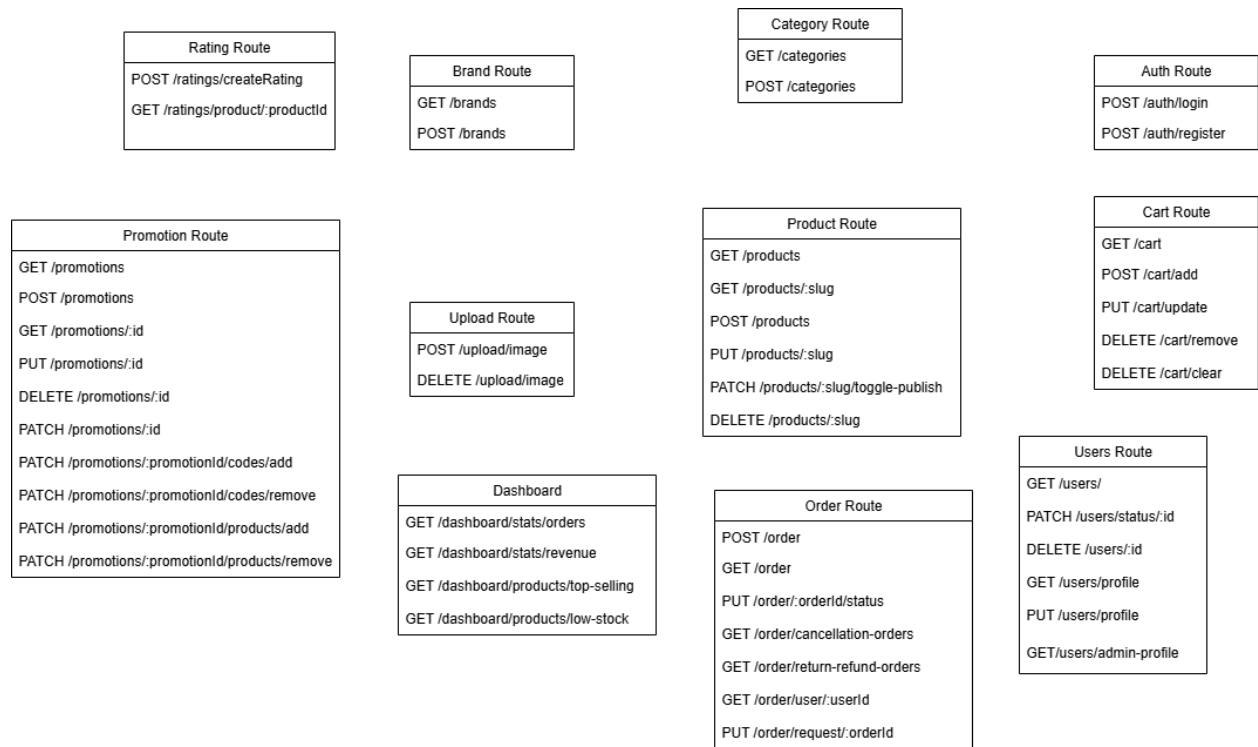
- Upload image: Upload images related to products or user profiles.
- Delete image: Delete uploaded images from the system.

Dashboard Controller: Manage revenue, order statistic and stock

- Get top selling product: Find best selling products based on sold orders
- Get low stock product: Find products that are running out of stock
- Order statistic: Count order with each state
- Revenue statistic: Calculate revenue over time

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4.5 Component: Router



The Router Component is to expose APIs from Back-End to Front-End which is retrieved from Services Component. The APIs are followed by the RESTful API (HTTP Method).

Products Route:

- GET Routes:
 - /products: get all products
 - /products/id: get information of a product by id
- POST Routes:
 - /products: create a product (require authorization)
- PUT Routes:
 - /products/:slug: edit a product by finding its slug (require authorization)
- PATCH Routes
 - /products/:slug/toggle-publish: handle hide/publish status of a product from store (require authorization)
- DELETE Routes:
 - /products/:slug: remove permanently a product from store (require authorization)

Brand Route:

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- GET Routes:
 - /brands: get all brands
- POST Routes:
 - /brands: create new brand (require authorization)

Category Route:

- GET Routes:
 - /categories: get all categories
- POST Routes:
 - /categories: create new category (require authorization)

Upload Route:

- POST Routes:
 - /upload/image: post new image (require authorization)
- DELETE Routes:
 - /upload/image: delete an image (require authorization)

Cart Route:

- GET Routes:
 - /cart: get cart information
- POST Routes:
 - /cart/add: add a product to cart
- PUT Routes:
 - /cart/update: update product in cart
- DELETE Routes:
 - /cart/remove: remove a product from cart
 - /cart/clear: remove all products from cart

Users Route:

- POST Routes:
 - /users/profile: get information about user's account
 - /users/admin-profile: get information about admin's account (require authorization)
- GET Routes:
 - /users/user-list: get all users (require authorization)
- PATCH Routes:
 - /users/status/:id: handle lock/unlock an account (require authorization)
- DELETE Routes:
 - /users/:id: delete permanently an account from system (require authorization)
- PUT Routes:
 - /users/profile: edit user's information

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Order Route:

- POST Routes:
 - /order: create an order
- GET Order:
 - /order: get all orders (require authorization)
 - /order/cancellation-orders: handle get all orders that are cancelled and requested to be cancelled (require authorization)
 - /order/return-refund-orders: handle get all orders that are requested to be exchanged or refunded (require authorization)
 - /order/user/:userID: get all orders from a user by user's id (require authorization)
- PUT Order:
 - /order/:orderId/status: handle changing status of order (require authorization)
 - /order/request/:orderId: handle send a request to refund, return for an order or cancel the order

Dashboard Route:

- GET Routes:
 - /stats/orders: get statistics of successfully delivered or cancelled orders (require authorization)
 - /stats/revenue: get sales revenue information (require authorization)
 - /stats/top-selling: get information about top-selling products (require authorization)
 - /stats/low-stock: get information about low-stock products (require authorization)

Authentication Route:

- POST Routes:
 - /auth/login: handle login to account
 - /auth/register: handle create new account

Promotion Route:

- GET Routes:
 - /promotions: get promotions list (require authorization)
 - /promotions/:id: get promotion detail by id (require authorization)
- POST Routes:
 - /promotions: create new promotion for products (require authorization)
- PUT Routes:
 - /promotions/:id: handle change information of promotion (require authorization)
- PATCH Routes:
 - /promotions/:id: handle active status of a promotion (require authorization)
 - /promotions/:promotionId/codes/add: add promotion code (require authorization)
 - /promotions/:promotionId/codes/remove: remove promotion code (require authorization)

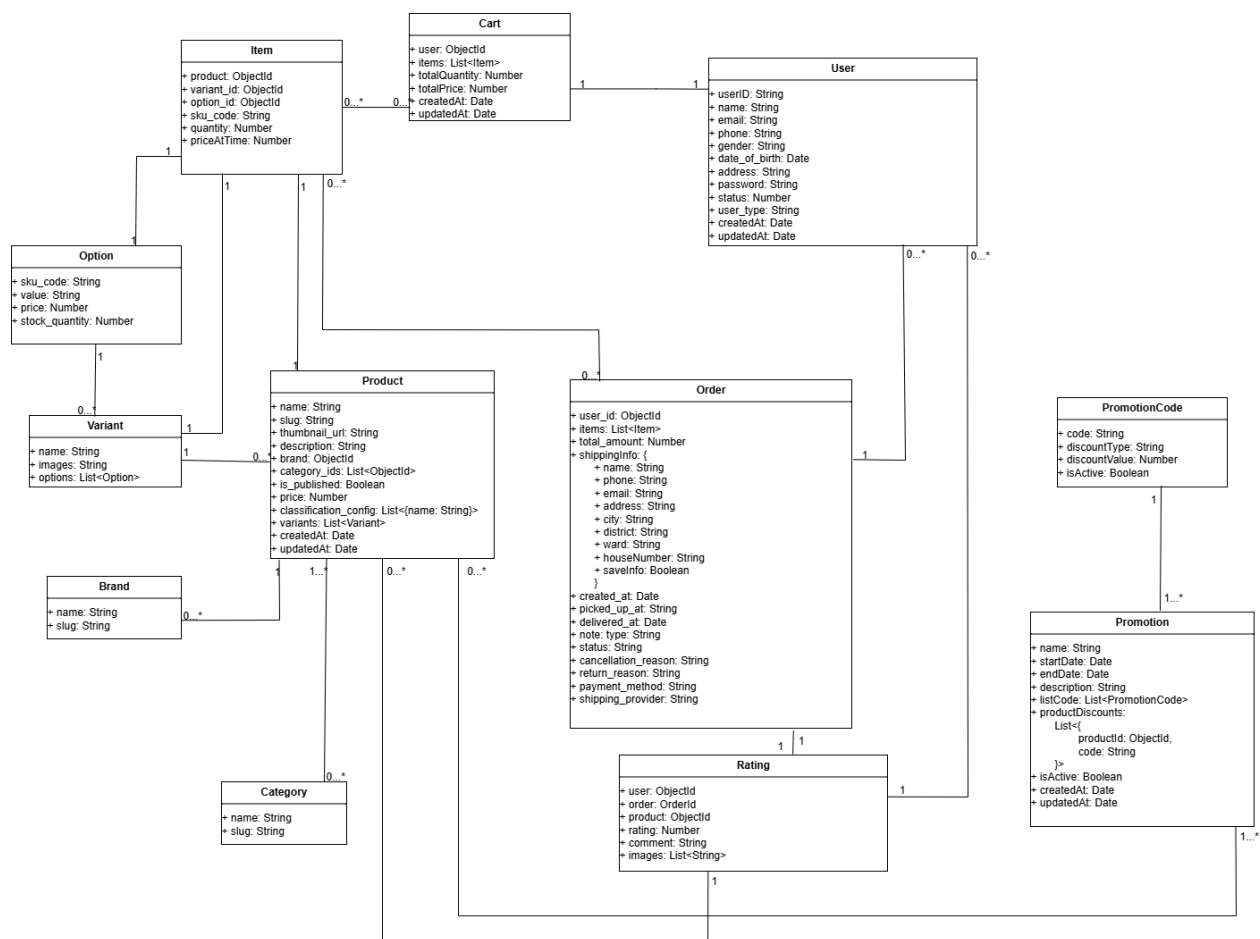
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- /promotions/:promotionId/product/add: add a product into promotion (require authorization)
- /promotions/:promotionId/product/remove: remove a product from promotion (require authorization)
- DELETE Routes:
 - /promotions/:id: delete a promotion (require authorization)

Rating Route:

- POST Routes:
 - /ratings/createRating: handle create comment and rating for a purchased product
- GET Routes:
 - /ratings/products/productId: handle get all comments and ratings of a product by product's id

4.6 Component: Models



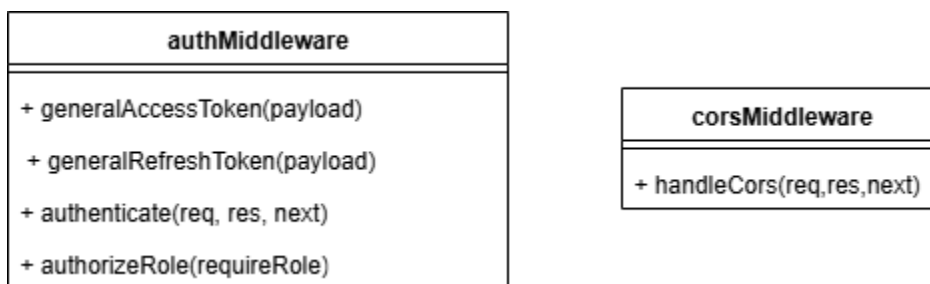
- **Product:** represents an item for sale, such as "Yonex Shoes". A product belongs to a single Brand (e.g., Yonex) and can be categorized under multiple Category (e.g., Shoes, Fashion). Each product may have

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multiple Variant, which represent different versions of the product, such as different colors (e.g., White, Red, Blue). Variant contains multiple Option, which describe specific configurations like size.

- **Brand:** represents a brand (e.g., Yonex, Asics), each Brand an own multiple Product.
- **Category:** represents a category of products (e.g., Shoes, Shirt, Racket), each Category can contain multiple Product.
- **Variant:** represents a specific classification of a Product (e.g., Yonex Shoes - Red). Each Variant belongs to only one Product, has a name (e.g., White, Red, Blue) and can have multiple Options, such as size or custom attributes.
- **Option:** represents a specific configuration inside a Variant, such as size. Each Option has a sku code, price, and stock quantity belongs to one variant only.
- **Item:** When a user selects a product for purchase or adds it to their cart, it becomes an Item which is a combination of product, variant, and option, along with the price at the time of selection. Items are components of both Cart and Order.
- **Cart:** represents the user's cart. Cart is associated with one User and holds multiple items prior to checkout.
- **Order:** represents the user's order. Order is a completed purchase and may contain many items.
- **User:** represents registered users, including customers and administrators. A user can manage their cart, place orders, and submit product Rating after purchase.
- **Rating:** each rating is tied to a specific order, product, and user, and include a number of star, comment, and optional images.
- **Promotion:** represents marketing campaigns or discount events that apply to specific products or are activated through promotion codes. Each promotion can be associated with multiple products via a list of product-specific discounts. Each Promotion includes multiple PromotionCodes, allowing more flexible discount strategies.

4.7 Component: Middleware



The **authMiddleware** is responsible for handling authentication and authorization within the application. It provides functionalities such as generating access and refresh tokens, authenticating user requests, and authorizing user roles.

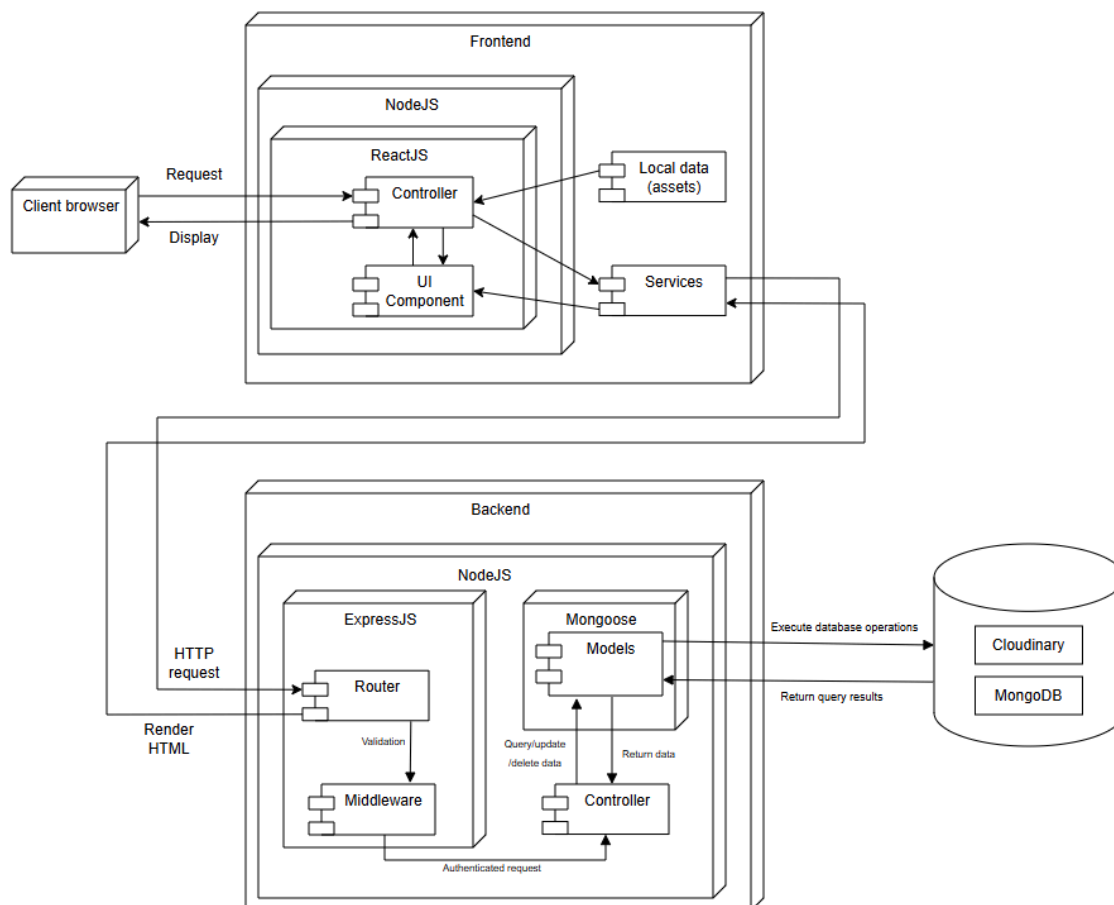
- **generalAccessToken(payload):** Handles the generation of the access token based on the provided payload.

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- **generalRefreshToken(payload)**: Responsible for generating a refresh token to enable the user to stay authenticated without having to log in again.
- **authenticate(req, res, next)**: Validates the authenticity of a user's request by checking their credentials.
- **authorizeRole(requireRole)**: Ensures that the user has the appropriate role required to access certain protected resources.

The **corsMiddleware** is responsible for handling cross-origin requests by allowing specific domains, HTTP methods, and headers to access the server's resources

5. Deployment



- **Client Browser:** This is where the user directly interacts with our website.
 - **Function:** It sends **Requests** (e.g., viewing a product, adding to cart) to the Frontend Node and

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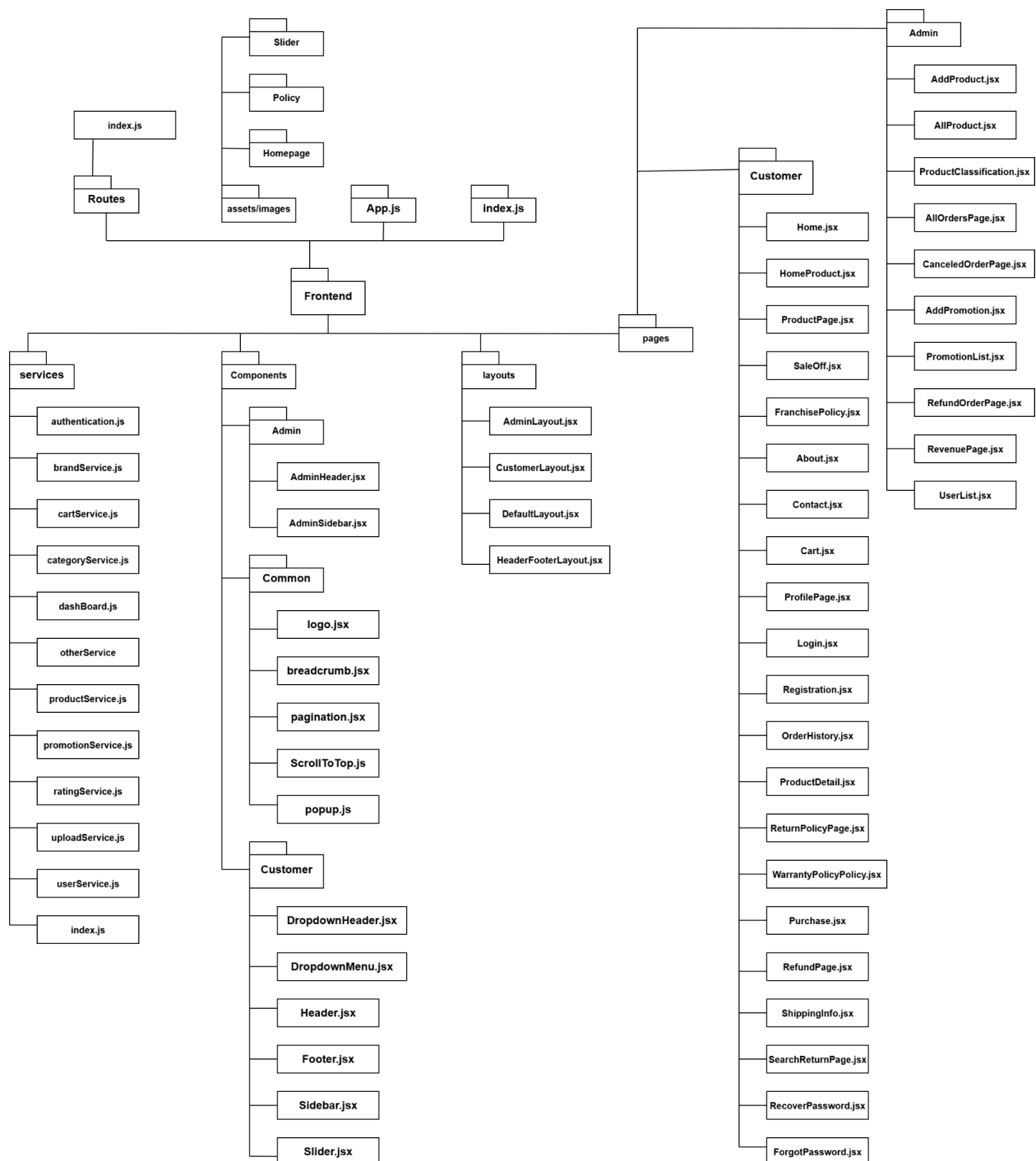
Displays the user interface (the website) returned by the frontend.

- **Frontend Node:** This is the client-side application running on NodeJS, responsible for everything the user sees and interacts with.
 - **Technology:** It uses **ReactJS** to build the User Interface (UI).
 - **Workflow:**
 - The **UI Components** and the **Controller** in ReactJS work together to create the visual interface.
 - When dynamic data is needed (like a product list or user information), the **Services** layer sends an HTTP request (an API call) to the Backend Node.
 - After receiving data from the backend, it updates the interface to display it to the user.
- **Backend Node:** This is the brain of the system, running on NodeJS and handling all business logic and data.
 - **Technology:** It uses the **ExpressJS** framework to build the API and **Mongoose** to interact with the database.
 - **Workflow:**
 - The **Router** receives the HTTP request from the Frontend Node.
 - The request passes through **Middleware** to handle tasks like authentication (checking if a user is logged in).
 - The **Controller** contains the core logic and processes the request (e.g., fetching a product from inventory, creating an order).
 - The Controller uses **Mongoose Models** to perform operations on the databases:
 - It stores/retrieves text-based data (product info, orders) from **MongoDB**.
 - It stores/retrieves media files (images, videos) from **Cloudinary**.
 - Finally, it returns the data (typically as JSON) to the Frontend Node.

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6. Implementation View

6.1 Front-end



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6.2 Back-end

